FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – IV

JEE (Advanced)-2019

PAPER – 2

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. A

Sol. In CM frame both the mass execute SHM with

$$\omega = \sqrt{\frac{k}{\mu}} = \sqrt{\frac{2k}{m}} SHM$$

Initially particles are at extreme

Distance = $L_0 + (L - L_0)\cos\sqrt{\frac{2k}{m}}t$

2. C

Sol. $$a=\left[\frac{F-(m+M)g\sin\theta}{(m+M)}\right]$$

So, $f=ma\cos\theta$

$$a=\left[\frac{F-(m+M)g\sin\theta}{(m+M)}\right]m\cos\theta$$

3. B

Sol. $a_B = 2m/s^2$ $(\uparrow)$

$$(\ell - x_3) + (x_2 - x_3) = k_1$$

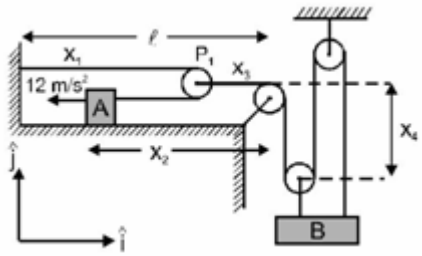
& $x_3 + 3x_4 = k_2$

$$a_2 - 2a_3 = 0 \Rightarrow a_3 = \frac{a_2}{2} = \frac{12}{2} = 6m/s^2$$

$[a_3 \Rightarrow$ acceleration of P_1 pulley$]$

$-a_3 + 3a_4 = 0 \quad \Rightarrow a_4 = \frac{a_3}{3} = \frac{6}{3} = 2m/s^2$

$[a_4 \Rightarrow$ acceleration of block B$]$



4. ABCD

Sol. The above lens mirror combination is an equivalent converging mirror.

5. AC

Sol. $f = mg\sin\theta$ and $PE\ \sin\theta = mg\sin\theta.R$

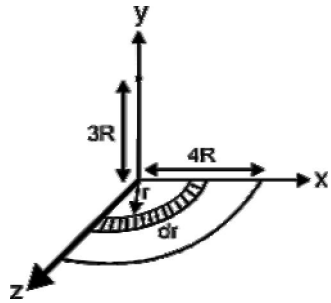
$q\ 2\ R\ E\sin\theta = mg\sin\theta\ R$

$E = \frac{mg}{2q}$

6. ABCD

Sol. (A) $v = \int_0^{4r} \frac{K\frac{\pi}{2} r dr\sigma}{\sqrt{r^2 + (3R)^2}} = \frac{\sigma R}{4\varepsilon_0}$

(B) $v = \int_0^{4r} \frac{K\frac{\pi}{2} r dr\sigma}{r} = \frac{\sigma R}{2\varepsilon_0}$

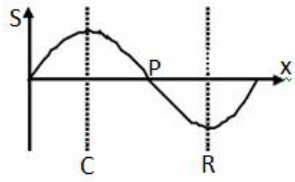


7. BD

Sol. External electric field and induced charge on outer surface of the sphere will produce a net zero field inside it. Net field at P must be vertically upward.

8. ACD

Sol. $P_C - P_R = 2BAK \qquad S = A\sin(kx - \omega t)$



9. D

10. A

Sol. (for Q. 9 to Q.10)

Let the initial temperature, pressure and volume of gas in 'A' be T_0, P_0, V_0 and the area of the position A and B be a, 2a. Now gas in chamber 'A' undergoes adiabatic compression whereas gas in chamber 'B' undergoes isothermal expansion.

Now solving for gas in 'A'

$P_{1A}V_{1A}^{\gamma} = P_{2A}V_{2A}^{\gamma} \Rightarrow P_0V_0^{\gamma} = P_{2A}(0.25V_0) \Rightarrow P_{2A} = 8P_0$

Solving for gas in 'B'

$P_{1B}V_{1B} = P_{2B}V_{2B} \Rightarrow P_{1B}V_0 = P_{2B} \times 2.5V_0 \Rightarrow P_{2B} = 0.4P_{1B}$

Also $(P_{2A} \times a) = (P_{2B} \times 2a)$ for gas is chambers 'B'

$\Rightarrow 8P_0a = P_2 \times 2a \Rightarrow P_2 = 4P_0$

$\Rightarrow 0.4\,P_1 = 4P_0 \Rightarrow P_1 = 10P_0$

Now comparing the moles of gas in A and B

$$n_A = \frac{P_0V_0}{RT_0}$$

$$n_B = \frac{10P_0V_0}{RT_0} = 10n_A$$

$m_B = 10m_A = 10\times 100\text{gm} = 1\text{kg}$

Again for gas is A, $\Delta Q = \Delta U + \Delta W$

$\Rightarrow 0 = \Delta U + \Delta W$

$$\Rightarrow 0 = \Delta U + \left\{\frac{P_0V_0 - 8P_0 \times \frac{V_0}{4}}{(\gamma - 1)}\right\}$$

$\Rightarrow \Delta U = 2P_0V_0 = 2\times 10^5 \times 10^{-2} = 2000\text{J}$

$F = 8P_0a = 8\times 10^5 \times 10^{-2} = 8000\text{N}$

SECTION – C

11. 7

Sol. $$P_0 + \frac{\rho v_1^2}{2} + 0 = P_A + \rho gh + \frac{\rho v_2^2}{2}$$

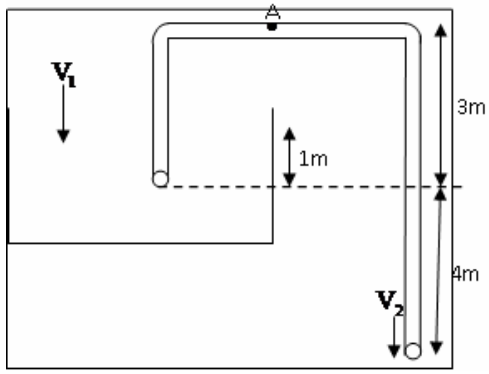
$\left(\because V_2 = \sqrt{2g\times 5} = 10 >> V_1\right)$

$$\left(\frac{\rho v_1^2}{2} - \frac{\rho v_2^2}{2}\right) - \rho gh = \left(P_A - P_0\right)$$

$$\left(P_A - P_0\right) = -\rho gh - \frac{\rho v_2^2}{2}$$

$$= -\frac{10^3}{2}\times 10\times 2 - \frac{10^3}{2}\times\frac{10\times 10}{1} = -\frac{7}{2}\times 10^4$$

$$= -\frac{7}{20}\times 10^5\,\text{N/m}^2 = -\frac{7}{20}\,atm$$



12. 2

Sol. The minima will be heard at P when a crest from S_1 and a trough from S_2 reach there at the same time. This will happen if $L_1 - L_2$ is $\lambda/2$ or $\lambda + (\lambda/2)$ or $2\lambda + (\lambda/2)$ and so on. Hence, the increase in L_1 between consecutive minima is λ_1 and from the data we see that $\lambda = 0.40\text{m}$.

Then $\lambda = v/f \Rightarrow f = 340/0.40 = 850\text{Hz}$.

13. 6

Sol. $X_{CM}=\dfrac{\int dm.x}{\int dm}=\dfrac{2L}{3}$

Mass of rod;

$M=\int_0^L \lambda_0 x\,dx=\dfrac{10^3}{2}$

Torque about 'O';

$F_B\dfrac{L}{2}\cos 37^\circ - Mg\dfrac{2L}{3}\cos 37^\circ = TL\cos 37^\circ$

$10^3\pi\left(\dfrac{1}{\pi}\right)\times 10\dfrac{1}{2}-\dfrac{10^4}{2}\times\dfrac{2}{3}=T$

$T=\dfrac{10^4}{6}N$

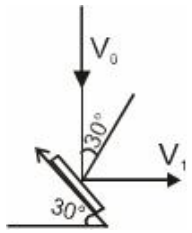
14. 1

Sol. $V_0\sin 30^\circ = V_1\cos 30^\circ \qquad (i)$

$eV_0\cos 30^\circ = V_1\sin 30^\circ \qquad (ii)$

Dividing (i) & (ii)

$\dfrac{1}{e}\tan 30^\circ=\dfrac{1}{\tan 30^\circ} \quad \dfrac{1}{e}=3$



15. 8

Sol. $f=2N$

$a_1 Kg=2m/s^2$

$S=\dfrac{1}{2}\times 2\times 4=4m$

$W_{friction}=-2\times 4=-8J$

16. 4

Sol. Since the rod is in translation so $\alpha=0$

$\therefore$ about O,

$F\dfrac{\ell}{2}\sin\theta - N\dfrac{\ell}{2}\cos\theta = 0$

$\Rightarrow F\sin\theta = N\cos\theta$

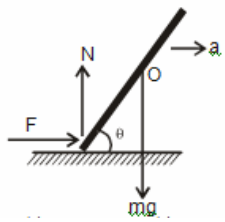
$\Rightarrow \tan\theta=\dfrac{N}{F}------(1)$

$Also, \quad N=Mg---(2)$

$So,\Rightarrow \tan\theta=\dfrac{Mg}{F}=\dfrac{40}{60}=\dfrac{2}{3}$

$\therefore \theta=\tan^{-1}\left(\dfrac{2}{3}\right)$

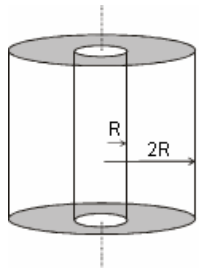
$X=3$



17. 4

Sol. $H = -K.2\pi r / \dfrac{dT}{dr}$

$$\int_{R_1}^{R_2} \frac{Hdr}{2\pi rl} = -K\int_{T_1}^{T_2} dT \Rightarrow H = \frac{2\pi lk\left(T_1 - T_2\right)}{\ln \dfrac{R_2}{R_1}} H_i = H_f$$



18. 2

Sol. $\dfrac{dN_A}{dt} = -\lambda_1 N_A, \dfrac{dN_B}{dt} = 2\lambda_1 N_A - \lambda_2 N_B$

$N_B = \text{maximum} \Rightarrow \dfrac{dN_B}{dt} = 0$

$\Rightarrow 2\lambda_1 N_A = \lambda_2 N_{B_{\max}} \Rightarrow N_{B_{\max}} = \dfrac{2\lambda_1}{\lambda_2} N_A$

$\Rightarrow N_{B_{\max}} = \dfrac{2\lambda_1}{\lambda_2} N_0 e^{-\lambda_1 t} = 2$

19. 3

Sol. T the FBD of any one rod is

$T = \mu N --------(1)$

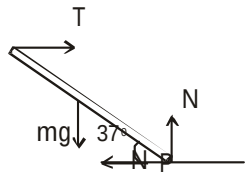
$mg = N --------(2)$

Taking torque of any one rod is

$\mathrm{mg}\dfrac{\mathrm{L}}{\mathrm{g}}\cos 37^{\circ} = \mathrm{TL}\sin 37^{\circ}$

$\mathrm{T} = \dfrac{\mathrm{mg}}{2}\dfrac{4}{3} = \dfrac{2\mathrm{mg}}{3}$

$\dfrac{2\mathrm{mg}}{3} = \mu\mathrm{mg} \Rightarrow \mu = \dfrac{2}{3}$



20. 6

Sol. $\dfrac{1}{\mathrm{f_L}} = \left(\dfrac{3}{2} - 1\right)\left(\dfrac{1}{20} - \dfrac{1}{20}\right)$

$\Rightarrow \mathrm{F}_L = 20cm$

$\dfrac{1}{\mathrm{f}} = \dfrac{1}{-10} - \dfrac{2}{20} = -\dfrac{1}{5}$

$\dfrac{1}{v} + \dfrac{1}{-30} = \dfrac{1}{-5} \Rightarrow \dfrac{1}{v} = \dfrac{1}{30} - \dfrac{1}{5}$

For lens $\dfrac{1}{v} - \dfrac{1}{-30} = \dfrac{1}{20}$

SECTION – D

21. 00025.00

Sol. $\overline{PQ} = \sqrt{15^2 + 20^2} = 25m$

$25 = \frac{V_P^2 \sin 2(45)}{g} \Rightarrow V_P = 5\sqrt{10} \text{m/s}$

$\therefore V_A^2 \sin^2\theta - 2\times 10\times 12.5 = \left(5\sqrt{5}\right)^2 \Rightarrow V_A \sin\theta = 5\sqrt{15} \text{m/s}$

$V_A \cos\theta = V_P \cos 45^o = 5\sqrt{5} \text{m/s} \text{ and } \theta = 60^o$

$\therefore \overline{AB} = \frac{V_A^2 \sin 2\theta}{g} = 25\sqrt{3}$

22. 00030.00

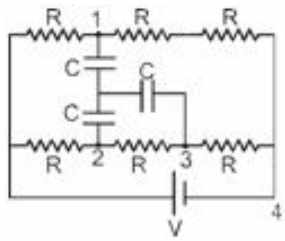
Sol. Consider equation of torque about C

$\left[mg\frac{4R}{3\pi} \right] = \frac{mR^2}{2}\alpha$

$\alpha = \frac{8g}{3\pi R} = \frac{8\pi g}{3\pi^2 R} = \frac{8\pi g}{30R}$

23. 00018.00

Sol. No current passes through capacitors in steady state. Assume potential at point '4' to be zero.

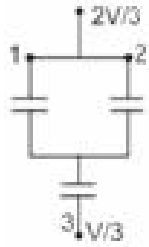


Then points '1' and '2' are at same potential $\frac{2V}{3}$

Hence $C_1 \text{and } C_2$ can be taken in parallel

The potential at point 3 is $\frac{V}{3}$

$\therefore$ Equivalent circuit of all three capacitors is shown Hence potential difference across capacitor C_3 is



$= \frac{2C}{2C+C} \times \left(\frac{2V}{3} - \frac{V}{3} \right) = \frac{2V}{9}$

Chemistry

PART – II

SECTION – A

24. B

Sol. NH_3 is a stronger field ligand than H_2O & causes greater crystal field splitting.

25. C

Sol. Radical are generally non-selective in their attack & readily combine with any species having unpaired electron & these can be produced by polar species like R – X in gas phase or in non polar solutions.

26. A

Sol. $A \rightleftharpoons B + C \qquad K_C = 10^6$

$1-x \quad x \quad x$

$\approx 0 \quad \approx 1M \quad \approx 1M$

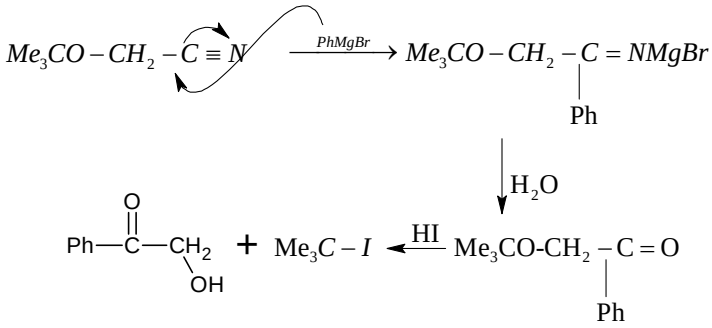
$B + \quad D \quad \rightleftharpoons A \qquad K_C = 10^{-6}$

$1-x \; 1-x \qquad x$

$\approx 1 \; \approx 1 \qquad \dfrac{x}{(1)(1)} = 10^{-6} \Rightarrow x = 10^{-6}$

27. CD

Sol.



28. ABD

Sol. First step involving formation of carbanion is slow & second is fast. Once the carbanion is formed, it undergoes rapid attack by an electrophile

29. ABC

Sol. $\Delta U_{BC} = nC_V(T_2 - T_1) = -750R, \; \Delta H_{CD} = nC_p(T_2 - T_1)$

$B \rightarrow C$

$\dfrac{P_B}{T_B} = \dfrac{P_C}{T_C} \Rightarrow P_B = 2\,bar$

$C \rightarrow D$

$nC_V(T_2 - T_1) = -P_{ex}(V_2 - V_1)$

$\Delta H_{ABCD} \neq 0$ as the whole process is not cyclic

$\Delta H_{AB} = \Delta U_{AB} = 0 \because \Delta T = 0$

30. ACD

Sol. $'A' is\ CrO_3$

$$CrO_3 + NaOH \longrightarrow Na_2CrO_4 + H_2O$$

$$yellow\ solution$$

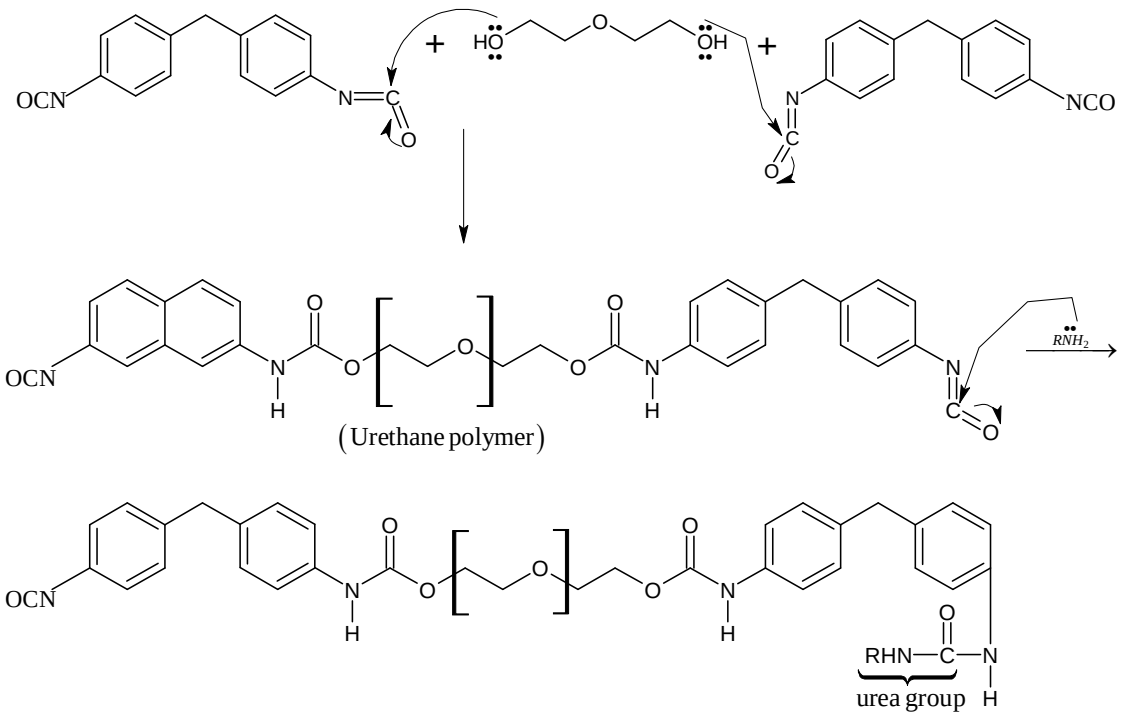
$$CrO_3 \xrightarrow{heat} Cr_2O_3 + O_2$$

31. ABD

Sol. Distance between nearest neighbours in BCC = Distance between test & octahedral void in FCC $= \sqrt{3}a / 4$ and distance between second nearest neighbours in both is a

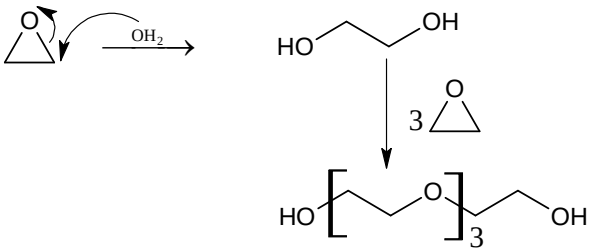
32. B

Sol.



33. C

Sol.



SECTION – C

34. 5

Sol. n=3

The subshell of this shell contains no radial node as described by wave function

So it is 3d .

35. 3

Sol. SOF_4 is sp^3d hybrid but contains no lone pair.

SeF_3^+ & $SeOCl_2$ contains lone pairs in sp^3 orbitals (no d - character)

XeF_3^+ is pentagonal bipyramidal with one lone pair.

BrF_6^- is perfectly octahedral containing lone pair in pure s-orbital

TeF_5 & ICl_4^- are sp^3d^2 hybrid with one & two lone pairs respectively.

36. 4

Sol. $CH_2O + O_2 \longrightarrow CO_2 + H_2O \ \Delta H_1 = -112 ----(1)$

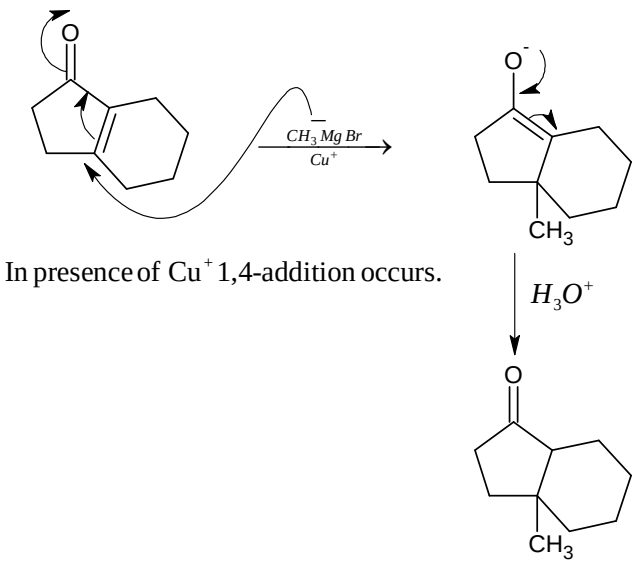
$nCH_2O + nO_2 \longrightarrow nCO_2 + nH_2O \ \Delta H_2 = -356 ---(2)$

$nCH_2O \longrightarrow (CH_2O)_n \ \Delta H_3 = -92\,k\,Cal ----(3)$

Equation (1)× n – equation (2) = equation (3)

37. 0

Sol.



In presence of Cu^+ 1,4-addition occurs.

38. 5

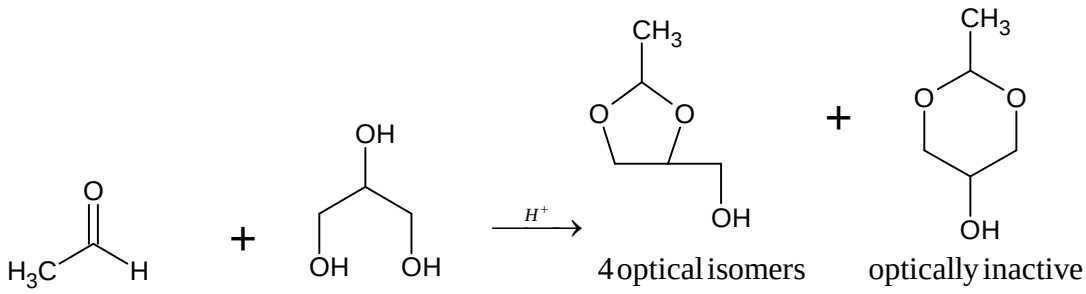
Sol. $XeF_4, NCl_3, PCl_3, TeF_6, SeBr_4$ are hydrolysed at room temperature but NF_3, CCl_4 & SF_6 are either not hydrolysed or require drastic conditions for hydrolysis.

39. 2

Sol. $(n_f)_{reaction} = 6$

40. 4

Sol.



4 optical isomers optically inactive

41. 2

Sol. Sugars differing around C_1 & C_2 only give same osazone.

42. 5

Sol. $r=k[O_3]^2$

$$-\frac{d[O_3]}{dt}=k[O_3]^2$$

$$=k\left(2\times10^{-8}\right)^2$$

43. 5

Sol. Soap solution, acidic sols, sulfide sols, metal sols & acidic dyes are negatively charged.

SECTION – D

44. 00026.10

Sol. $2H_2O+2e\xrightarrow{electrolysis}H_2+2OH^-$

$$\text{millimoles}\,of\ OH^-\ produced=\frac{50\times8685}{96500}=4.5$$

Milimoles of acid present $=2.25$

Mass present $=0.261g$

Percentage $=26.100$

45. 05340.00

Sol. $\left[H_3O^+\right]=\sqrt{CK_a}$

Milligram of $AlCl_3=C\times0.4\times M_{AlCl_3}\times10^3$

46. 00000.06

Sol. $H^+\left(HA_1\right)\longrightarrow H^+\left(HA_2\right)\&\quad H^+=\sqrt{Ka.C}$

$$E_{cell}=0-\frac{0.06}{1}\log\frac{\sqrt{10^{-5}.C}}{\sqrt{10^{-3}.C}}$$

Mathematics

PART – III

SECTION – A

47. A

Sol. $\because P \equiv (a\cos\theta, b\sin\theta)$

And $Q \equiv \left(a\cos\left(\frac{\pi}{2}+\theta\right), b\sin\left(\frac{\pi}{2}+\theta\right)\right)$

$Q \equiv (-a\sin\theta, b\cos\theta)$

Let midpoint of PQ is (x, y), then

$\frac{2x}{a} = (\cos\theta - \sin\theta)$

$\frac{2y}{b} = (\sin\theta + \cos\theta)$

Squaring and adding equation (iii) and equation (iv), we get

$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{1}{2}$

48. B

Sol. Since $A.M. \geq G.M.$

$\therefore \sin A + \sin B \geq 2\sqrt{\sin A.\sin B}$

Similarly, $\sin B + \sin C \geq 2\sqrt{\sin B.\sin C}$

And $\sin C + \sin A \geq 2\sqrt{\sin C.\sin A}$

$\Rightarrow (\sin A + \sin B)(\sin B + \sin C)$

$(\sin C + \sin A) \geq 8\sin A.\sin B.\sin C$

49. D

Sol. $\overline{\alpha} = \frac{\overline{z} - \omega}{k^2 + \overline{z}\omega}$

But $z\overline{z} = \omega\overline{\omega} = k^2, thus$

$\overline{\alpha} = \frac{\frac{k^2}{z} - \frac{k^2}{\overline{\omega}}}{k^2 + \frac{k^2}{z}\frac{k^2}{\overline{\omega}}} = \frac{\overline{\omega} - z}{z\overline{\omega} + k^2} = -\alpha$

$\Rightarrow \alpha + \overline{\alpha} = 0$

$\Rightarrow \mathrm{Re}(\alpha) = 0$

50. A

Sol. Both the lines passes through the origin. Line L_1 is parallel to the vector $\vec{V}_1$.

$\vec{V}_1 = \left(\cos\theta + \sqrt{3}\right)\hat{i} + \left(\sqrt{2}\sin\theta\right)\hat{j}\left(\cos\theta - \sqrt{3}\right)\hat{k}$

And L_2 is parallel to the vector $\vec{V}_2$.

$$\therefore \cos\alpha = \frac{\vec{V}_1.\vec{V}_2}{\left|\vec{V}_1\right|\left|\vec{V}_2\right|} = \frac{a\left(\cos\theta+\sqrt{3}\right)+\left(b+\sqrt{2}\right)\sin\theta+c\left(\cos\theta-\sqrt{3}\right)}{\sqrt{a^2+b^2+c^2}\sqrt{(\cos\theta+\sqrt{3})^2+2\sin^2\theta(\cos\theta+\sqrt{3})}}$$

$$= \frac{(a+c)\cos\theta+b\sqrt{2}\sin\theta+(a-c)\sqrt{3}}{\sqrt{a^2+b^2+c^2}\sqrt{2+6}}$$

For $\cos\alpha$ to be independent of θ, we get

$a + c = 0$ and $b = 0$

$$\therefore \cos\alpha = \frac{2a\sqrt{3}}{a\sqrt{2}.2\sqrt{2}} = \frac{\sqrt{3}}{2} \text{ or } \alpha = \frac{\pi}{6}$$

51. ABD

Sol. From the graph it is clear that $a < 0$ and $c > 0$.

Now since vertex is in first quadrant $-\frac{b}{2a} > 0$.

$-b < 0 \Rightarrow b > 0$

52. BC

Sol. $\left|\frac{1}{z_2}+\frac{1}{z_1}\right| = \left|\frac{1}{z_2}-\frac{1}{z_1}\right| \Rightarrow |z_1+z_2| = |z_1-z_2|$

$\Rightarrow \cos(\theta_1 - \theta_2) = 0$ where θ_1 and θ_2 are arguments of z_1 and z_2.

53. BCD

Sol. Equating coefficients of x, x^2 and x^3 both sides we get,

$a_1 = n$, $a_2 = \frac{n(n-1)}{2} + 2$, $a_3 = \frac{n(n-1)(n-2)}{6} + 2n$.

Now $2a_2 = a_1 + a_3$ gives $n^3 - 9n^2 + 26n - 24 = 0$

54. AC

Sol. f(x) is a decreasing function and for the major axis to be the x-axis.

$f(k^2 + 2k + 5) > f(k + 11)$

or $k^2 + 2k + 5 < k + 11$

or $k \in (-3, 2)$

Then for the remaining values of k, i.e., $k \in (-\infty, -3) \cup (2, \infty)$ k, the major axis is the y-axis.

55. C

Sol. $$\frac{{}^5C_1 + \frac{1}{2}\times {}^2C_1}{10} = \frac{3}{5}$$

56. B

Sol. $$p(F/H) = \frac{P(F)P(H/F)}{\sum P(Hi)\times P(Hi/F)}$$

$F \rightarrow$ fair coin

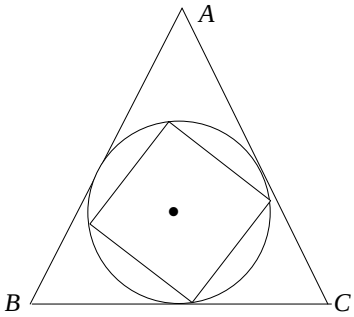
$H \rightarrow$ head appears

$$P(H/F) = \frac{\frac{1}{5}\times\frac{1}{2}}{\frac{1}{5}\times\frac{1}{2}+\frac{4}{5}\times\frac{5}{8}} = \frac{1}{6}$$

SECTION – C

57. 6

Sol.



$$R = \frac{a}{2\sin A} = \frac{P}{\sqrt{3}}$$

$$r = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2} = \frac{R}{2}$$

$$\Rightarrow r = \frac{P}{2\sqrt{3}}$$

$$\Rightarrow \text{ side of } = r\sqrt{2} = \frac{P}{\sqrt{6}}$$

Area = $\frac{P^2}{6} \Rightarrow K = 6$

58. 0

Sol. $f(x) + 2f(1-x) = x^2 + 1 \quad .....(1)$

Replace x by (1-x)

$f(1-x) + 2f(x) \neq (1-x)^2 + 1 \quad ....(2)$

(2) – (1)

$3f(x) = 2(1-x)^2 + 2 - x^2 - 1$

$f(3) = 0$

59. 1

Sol. Let equation of line along minor axis is

$lx + my + n = 0$

$$\frac{(x-\alpha)^2}{a^2} + \frac{(y-\beta)^2}{b^2} = 1$$

$l\alpha + m\beta + n = 0$

$\Rightarrow$ only one arbitary constant will be left (size is same)

$\Rightarrow$ degree of differential equation will be '1'

60. 9

Sol. $\left(1 + x + x^2 + x^3\right)^{-3}$

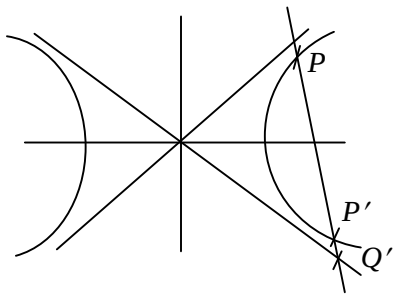
$$=\left(\frac{1-x^4}{1-x}\right)^{-3}=\left(1-x^4\right)^{-3}\left(1-x\right)^{-1}$$

Coefficient of x^r in expansion of $\left(1-x\right)^{-n}$ is ${}^{n+r-1}C_r$

61. 2

Sol. For any line in case of hyperbola we know that $PQ = P'Q'$

$$\frac{PQ}{P'Q'}+\frac{PQ'}{P'Q}=1+\frac{PP'+P'Q'}{PP'+PQ}=2$$



62. 8

Sol. $$=\begin{bmatrix}3 & -4\\ 1 & -1\end{bmatrix}^k=\left(\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}+\begin{bmatrix}2 & -4\\ 1 & -2\end{bmatrix}\right)^k$$

$$=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}+k\begin{bmatrix}2 & -4\\ 1 & -2\end{bmatrix}\quad\left(\text{as}\begin{bmatrix}2 & -4\\ 1 & -2\end{bmatrix}^2=0\right)$$

$$=\begin{bmatrix}1+2k & -4k\\ k & 1-2k\end{bmatrix}$$

$$\Rightarrow \sum_{k=1}^{n}A^k=\begin{bmatrix}n+n(n+1) & -2n(n+1)\\ \dfrac{n(n+1)}{2} & n-n(n+1)\end{bmatrix}$$

$$\Rightarrow \text{Let}\left(\sum_{k=1}^{n}A^k\right)=n^2=64$$

$$\Rightarrow n=8$$

63. 8

Sol. $\left[x/9\right]=\left[x/11\right]=I$

$\Rightarrow I \le x/9 < I+1,\ I \le x/11 < I+1,$ Solution is possible only when $11I < 9I+9$

$$\Rightarrow 0 \le I < \frac{9}{2}$$

Total number of solution $\sum_{I=0}^{4}(9\text{-}2I)\text{-}1=24$

64. 2

Sol. $f(x)=2-x \quad x \le 0$

$2+x \quad x>0$

Hence minimum value of $f(x)=2$

65. 3

Sol. $f(n+1)-f(n)=\frac{1}{\pi}\int_0^{\pi/2}\left(\frac{\sin^2(n+1)\theta-\sin^2 n\theta}{\sin^2\theta}\right)d\theta$

$=\frac{1}{\pi}\int_0^{\pi/2}\frac{\sin(2n+1)\theta}{\sin\theta}d\theta=\frac{1}{\pi}\left(\int_0^{\pi/2}\frac{\sin 2n\theta\cos\theta d\theta}{\sin\theta}+\int_0^{\pi/2}\cos 2n\theta\, d\theta\right)$

$=\frac{1}{\pi}\int_0^{\pi/2}\frac{\sin 2n\theta\cos\theta d\theta}{\sin\theta}=\frac{1}{\pi}2\int_0^{\pi/2}(\cos\theta+\cos 3\theta+\ldots\cos(2n-1)\theta)\cos\theta d\theta$

$=\frac{1}{\pi}\int_0^{\pi/2}(1+\cos 2\theta)+(\cos 2\theta+\cos 4\theta)+\ldots+(\cos(2n-2)\theta+\cos 2n\theta)d\theta$

$=\frac{1}{\pi}\int_0^{\pi/2}d\theta=\frac{1}{2}$

$\Rightarrow f(n+1)-f(n)=\frac{1}{2}$

$\Rightarrow f(n)=\frac{n}{2}$ (as f(1)=$\frac{1}{2}$)

Hence, $\frac{f(15+f(3)}{f(15)-f(9)}$ = 3

66. 3

Sol. General term $T_r=\frac{2r+3}{3^{r+1}.r(r+1)}$

$\Rightarrow T_r=\frac{1}{3^r.r}-\frac{1}{3^{r+1}.(r+1)}$

$\Rightarrow T_1=\frac{1}{3\times1}-\frac{1}{3^2\times2}$

$\Rightarrow T_2=\frac{1}{3^2\times2}-\frac{1}{3^3\times3}$

$\Rightarrow T_n=\frac{1}{3^n\times n}-\frac{1}{3^{n+1}\times(n+1)}$

Let $S_n = T_1 + T_2 + \ldots\ldots\ldots\ldots + T_n = \frac{1}{3\times1}-\frac{1}{3^{n+1}\times(n+1)}$

$\lim_{x\to\infty} S_n=\frac{1}{3}-0 \quad \Rightarrow \frac{k}{9}=\frac{1}{3} \quad \Rightarrow k=3$

SECTION – D

67. 00003.50

Sol. The function $u=f(x)=\frac{1}{x-1}$ is discontinuous at the point $x=1$. The function

$y=g(u)=\frac{1}{u^2+u-2}=\frac{1}{(u+2)(u-1)}$ is discontinuous at $u=-2$ and $u=1$

When $u=-2, \frac{1}{x-1}=u=-2 \Rightarrow x=\frac{1}{2}$

When $u=1, \frac{1}{x-1}=1 \Rightarrow x=2$. Hence, the composite function $y=g\left(f\left(x\right)\right)$ is discontinuous at three points $x=1/2,1,2$.

68. 00011.40

Sol. Let $\mathrm{I}\left(b\right)=\int_0^1 \frac{x^b-1}{\ln x}dx \Rightarrow \mathrm{I}'\left(b\right)=\int_0^1 \frac{x^b \ln x}{\ln x}dx=\int_0^1 x^b dx$

Or $\mathrm{I}'\left(b\right)=\left[\frac{x^{b+1}}{b+1}\right]_0^1=\frac{1}{b+1} \Rightarrow \mathrm{I}\left(b\right)=\int \frac{db}{b+1}=\ln\left(b+1\right)+c$

If $b=0$, then $I\left(b\right)=0 \Rightarrow c=0$. Hence, $\mathrm{I}\left(\mathrm{b}\right)=\ln\left(b+1\right)$

69. 00265.00

Sol. The number of ways when number student falled in any examination $=(2^3\text{-}1)^3$

The number of ways when out of above cases atleast are subject was not cleaned by any students $=3C_1(3)^3$

The number of ways when out of above cases atleast any two subjects were not cleaned by any student $=3C_2(1)^3$

So required cases $=(2^3-1)^3-3C_1 3^3+3C_2$ = 265